YE LIU (刘 烨)

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Professional Experience

The Ohio State University (OSU) Columbus, USA
Postdoctoral Scholar, Advisor: Prof. Xia Ning *Sep. 2025 - now*
- Job Description: Agentic AI and large language models (LLMs), with an emphasis on science and healthcare.

ByteDance - AI LAB Beijing, China
Algorithm Intern *Feb. 2023 - Aug. 2023*
- Job Description: RLHF for Large Language Models, especially focusing on SFT and Reward Model.

Research Interest

My research focuses on agentic artificial intelligence built upon large language models (LLMs), with the goal of developing adaptive, reliable, and collaborative AI systems for complex real-world domains, particularly healthcare and scientific discovery. My research interests include:

- **Tool Learning and Agentic AI**: Enabling LLM-based agents to understand, select, and utilize diverse external tools through representation learning, continual adaptation, and human-in-the-loop feedback.
- **Trustworthy Large Language Models**: Improving the reasoning reliability and decision quality of LLMs via retrieval-augmented generation, instruction tuning, and interaction with structured knowledge and external systems, addressing the high-stakes and low-tolerance requirements of health and science applications.
- **System-Level AI for Healthcare and Science**: Designing end-to-end agentic AI pipelines that integrate perception, retrieval, reasoning, and action for scientific discovery and healthcare decision-making.

Education

University of Science and Technology of China (USTC), Advisor: Prof. Enhong Chen Hefei, China
Ph.D., in Data Science - Computer Science and Technology, GPA: 3.90/4.3 *Sep. 2019 - June 2025*
Hong Kong University of Science and Technology (HKUST), Advisor: Prof. Xiaofang Zhou HK, China
Visiting Ph.D. Student, in Computer Science and Engineering *May 2024 - Nov. 2024*
University of Science and Technology of China (USTC) Hefei, China
B.E., in Electronic Information Engineering, GPA: 3.78/4.3 *Sep. 2015 - June 2019*

Honors

- 2025, Outstanding Graduate of USTC and Anhui Province
- 2023, CICAI Finalist of Best Paper Award (Top-3)
- 2016, National Scholarship

Teaching

- VETPREV 8725: Quantitative Methods Capstone Seminar (Spring 2026, OSU)
- COMP6110P.02: Machine Learning and Knowledge Discovery (Fall 2021, USTC)
- CS1001A.13: Computer Programming A (Fall 2020, USTC)
- 011X3001: Introduction to Data Science (Fall 2019, USTC)
- 011164.01: Computer Programming I (Fall 2018, USTC)

Academic Service

- **Area Chair:** ACL Rolling Review.
- **Conference Reviewer:** ACL Rolling Review, NeurIPS, ICLR, CVPR, WWW, AAAI, ECCV, SIGIR, COLM.
- **Journal Reviewer:** IEEE TKDE, IEEE TBD, Neural Networks, Information Sciences, JCST.

Mentoring

- **Xukai Liu**: USTC UG → USTC Ph.D. Worked on knowledge alignment and linking (Findings of ACL 2023, EMNLP 2024 Main).
- **Yanghai Zhang**: USTC UG → USTC Ph.D. Worked on knowledge-enhanced multimodal summarization (Findings of ACL 2024).
- **Haoyu Tang**: USTC UG → Renmin University of China M.S. Machine unlearning for LLMs (ICDM 2025).
- **Xinyi Ling**: M.S. student at OSU. Worked on conversational recommendation and LLM agents (COLM 2026).

Reference

- **Dr. Enhong Chen:**
  Professor at USTC. Ph.D. Advisor. Email: cheneh@ustc.edu.cn.
- **Dr. Xiaofang Zhou:**
  Professor at HKUST. Advisor of my visit in HKUST. Email: zxf@ust.hk.
- **Dr. Xia Ning:**
  Professor at OSU. Postdoctoral Advisor. Email: ning.104@osu.edu.

Preprints *(† Equal contribution)*

[1]. **Ye Liu**†, Reza Averly†, Avery Meyer, Frazier N. Baker, Ruoxi Gao, Daniel Adu-Ampratwum, Xia Ning. MADDIE: Multi-Agent Drug Discovery Engine. ArXiv 2026.

[2]. **Ye Liu**, Botao Yu, Xinyi Ling, Daniel Adu-Ampratwum, Xia Ning. ARMOR: An Agentic Framework for Reaction Feasibility Prediction via Adaptive Utility-aware Multi-tool Reasoning. ArXiv 2026.

[3]. Xukai Liu, **Ye Liu**, Shiwen Wu, Yanghai Zhang, Yihao Yuan, Kai Zhang, Qi Liu. Knowş-RAG: A Knowledge-aware RAG Framework with Adaptive Retrieval, Generation, and Filtering. ArXiv 2025.

[4]. Jiajun Zhu, **Ye Liu**, Meikai Bao, Kai Zhang, Yanghai Zhang, Qi Liu. Self-Reflective Planning with Knowledge Graphs: Enhancing LLM Reasoning Reliability for Question Answering. ArXiv 2025.

[5]. Zheng Huang, Xukai Liu, Tianyu Hu, Kai Zhang, **Ye Liu**. PPTBench: Towards Holistic Evaluation of Large Language Models for PowerPoint Layout and Design Understanding. ArXiv 2025.

[6]. Linan Yue, Yichao Du, Yizhi Wang, Weibo Gao, Fangzhou Yao, Li Wang, **Ye Liu**, Ziyu Xu, Qi Liu, Shimin Di, Min-Ling Zhang. Don't Overthink It: A Survey of Efficient R1-style Large Reasoning Models. ArXiv 2025.

Publications *(† Equal contribution)*

[1]. Xukai Liu†, **Ye Liu**†, Jipeng Zhang, Yanghai Zhang, Kai Zhang, Qi Liu. Layer-Order Inversion: Rethinking Latent Multi-Hop Reasoning in Large Language Models. The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.

[2]. Xinyi Ling†, **Ye Liu**†, Reza Averly, Xia Ning. Uncertainty as a Planning Signal: Multi-Turn Decision Making for Goal-Oriented Conversation. The 3rd Conference on Language Modeling (COLM), 2026.

[3]. **Ye Liu**, Jiajun Zhu, Kai Zhang, Haoyu Tang, Qi Liu, Enhong Chen. Detect, Investigate, Judge and Determine: A Novel LLM-based Framework for Few-shot Fake News Detection. The 25th IEEE International Conference on Data Mining (ICDM), 2025.

[4]. Haoyu Tang†, **Ye Liu**†, Xukai Liu, Kai Zhang, Yanghai Zhang, Qi Liu, Enhong Chen. Learn while Unlearn: An Iterative Unlearning Framework for Generative Language Models. The 25th IEEE International Conference on Data Mining (ICDM), 2025.

[5]. **Ye Liu**, Kai Zhang, Aoran Gan, Linan Yue, Feng Hu, Qi Liu, Enhong Chen. Empowering Few-Shot Relation Extraction with The Integration of Traditional RE Methods and Large Language Models. The 29th International Conference on Database Systems for Advanced Applications (DASFAA), 2024.

[6]. **Ye Liu**, Kai Zhang, Zhenya Huang, Kehang Wang, Yanghai Zhang, Qi Liu, Enhong Chen. Enhancing Hierarchical Text Classification through Knowledge Graph Integration. Findings of the 61st annual meeting of the Association for Computational Linguistics (ACL-Findings), 2023.

[7]. **Ye Liu**, Han Wu, Zhenya Huang, Hao Wang, Yuting Ning, Jianhui Ma, Qi Liu, Enhong Chen. TechPat: Technical Phrase Extraction for Patent Mining. ACM Transactions on Knowledge Discovery from Data (ACM TKDD), 2023.

[8]. **Ye Liu**, Han Wu, Zhenya Huang, Hao Wang, Jianhui Ma, Qi Liu, Enhong Chen, Hanqing Tao and Ke Rui. Technical Phrase Extraction for Patent Mining: A Multi-level Approach. The 2020 IEEE International Conference on Data Mining (ICDM), 2020.

[9]. Aoran Gan, **Ye Liu**, Hongbo Gang, Kai Zhang, Qi Liu, Guoping Hu. From Memorization to Discovery: A Novel Benchmark for Relational Triple Extraction. The 30th International Conference on Database Systems for Advanced Applications (DASFAA), 2025.

[10]. Feng Hu, Kai Zhang, **Ye Liu**, Meikai Bao, Xukai Liu, Junyu Lu, Linbo Zhu, Qi Liu. Learnable Relational Knowledge Distillation For Language Model Compression. The 30th International Conference on Database Systems for Advanced Applications (DASFAA), 2025.

[11]. Linan Yue, Qi Liu, **Ye Liu**, Weibo Gao, et al. Learning from shortcut: a shortcut-guided approach for explainable graph learning. Frontiers of Computer Science (FCS), 2025.

[12]. Yanghai Zhang, **Ye Liu**, Shiwei Wu, Kai Zhang, Xukai Liu, Qi Liu, Enhong Chen. Leveraging Entity Information for Cross-Modality Correlation Learning: The Entity-Guided Multimodal Summarization. Findings of the 62nd annual meeting of the Association for Computational Linguistics (ACL-Findings), 2024.

[13]. Xukai Liu, **Ye Liu**, Kai Zhang, Kehang Wang, Qi Liu, Enhong Chen. OneNet: A Fine-Tuning Free Framework for Few-Shot Entity Linking via Large Language Model Prompting. The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024.

[14]. Linan Yue, Qi Liu, **Ye Liu**, Weibo Gao, et al. Cooperative Classification and Rationalization for Graph Generalization. The ACM Web Conference, 2024 (WWW), 2024.

[15]. Linan Yue, Qi Liu, Weibo Gao, **Ye Liu**, Kai Zhang, Yichao Du, Li Wang, et al. Federated Self-Explaining GNNs with Anti-shortcut Augmentations. The 41st International Conference on Machine Learning (ICML), 2024.

[16]. Kehang Wang, **Ye Liu**, Kai Zhang, Qi Liu, Yankun Ren, Xinxing Yang, Longfei Li, Jun Zhou. QoMRC: Query-oriented Machine Reading Comprehension Framework for Aspect Sentiment Triplet Extraction. The 29th International Conference on Database Systems for Advanced Applications (DASFAA), 2024.

[17]. Zhijun Dong, Likang Wu, Kai Zhang, **Ye Liu**, Yanghai Zhang, Zhi Li, Hongke Zhao and Enhong Chen. FZR: Enhancing Knowledge Transfer via Shared Factors Composition in Zero-Shot Relational Learning. The 33rd ACM International Conference on Information and Knowledge Management (CIKM), 2024

[18]. Linan Yue, Qi Liu, Yichao Du, Weibo Gao, **Ye Liu**, et al. FedJudge: Federated Legal Large Language Model. The 29th International Conference on Database Systems for Advanced Applications (DASFAA), 2024.

[19]. Xukai Liu, Kai Zhang, **Ye Liu**, Enhong Chen, Zhenya Huang, , Linan Yue, Jiaxian Yan. RHGH: Relation-gated Heterogeneous Graph Network for Entity Alignment in Knowledge Graphs. Findings of the 61st annual meeting of the Association for Computational Linguistics (ACL-Findings), 2023.

[20]. Meikai Bao, Qi Liu, Kai Zhang, **Ye Liu**, Linan Yue, longfei li, Jun Zhou. Keep Skills in Mind: Understanding and Implementing Skills in Commonsense Question Answering. The 32nd International Joint Conference on Artificial Intelligence (IJCAI), 2023.

[21]. Kehang Wang, Qi Liu, Kai Zhang, **Ye Liu**, Hanqin Tao, Zhenya Huang, Enhong Chen. Class-Dynamic and Hierarchy-Constrained Network for Entity Linking. The 28th International Conference on Database Systems for Advanced Applications (DASFAA), 2023.

[22]. Xin Jin, Qi Liu, Linan Yue, **Ye Liu**, Lili Zhao, Weibo Gao, Zheng Gong, Kai Zhang, Haoyang Bi. Diagnosis then Aggregation: An Adaptive Ensemble strategy for Keyphrase Extraction. CAAI International Conference on Artificial Intelligence (CICAI), 2023. (Finalist of Best Paper Award!)

[23]. Huijie Liu, Han Wu, Le Zhang, Runlong Yu, **Ye Liu**, Chunli Liu, Minglei Li, Qi Liu, Enhong Chen. A Hierarchical Interactive Multi-channel Graph Neural Network for Technological Knowledge Flow Forecasting. Knowledge and Information Systems (KAIS), 2022.

[24]. Guanqi Zhu, Hanqing Tao, Han Wu, Liyi Chen, **Ye Liu**, Qi Liu, Enhong Chen. Text Classification via Learning Semantic Dependency and Association. The 2022 IEEE World Congress on Computational Intelligence - IJCNN (IJCNN), 2022.

[25]. Yixiao Ma, Shiwei Tong, **Ye Liu**, Likang Wu, Qi Liu, Enhong Chen, Wei Tong, Zi Yan. Enhanced Representation Learning for Examination Papers with Hierarchical Document Structure. The 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2021.

[26]. Songtao Fang, Zhenya Huang, Ming He, Shiwei Tong, Xiaoqing Huang, **Ye Liu**, Jie Huang, Qi Liu. Guided Attention Network for Concept Extraction. The 30th International Joint Conference on Artificial Intelligence (IJCAI), 2021.

[27]. Huijie Liu, Han Wu, Le Zhang, Runlong Yu, **Ye Liu**, Chunli Liu, Qi Liu, Enhong Chen. Technological Knowledge Flow Forecasting through A Hierarchical Interactive Graph Neural Network. IEEE International Conference on Data Mining (ICDM), 2021.

[28]. Yuting Ning, **Ye Liu**, Zhenya Huang, Haoyang Bi, Qi Liu, Enhong Chen, Dan Zhang. Stable and Diverse: A Unified Approach for Computerized Adaptive Testing. The 7th IEEE International Conference on Cloud Computing and Intelligence Systems (CCIS), 2021.

[29]. Lili Zhao, Linan Yue, Yanqing An, **Ye Liu**, Kai Zhang, Weidong He, Yanmin Chen, Qi Liu. Legal Judgment Prediction with Multiple Perspectives on Civil Cases. The 1st CAAI International Conference on Artificial Intelligence (CICAI), 2021.